

# DAX Measures Documentation

## Cargo Operations Analytics Dashboard

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### 1. Total Containers

#### Purpose

Calculates the total number of containers moved.

```
Total Containers =  
SUM(  
fact_cargo_movements[container_count]  
)
```

Used In:

- KPI Cards
  - Trend Analysis
  - Dashboard Summary
- 

### 2. Total Movements

#### Purpose

Counts total cargo movement transactions.

```
Total Movements =  
COUNT(  
fact_cargo_movements[movement_id]  
)
```

Used In:

- KPI Cards
  - Operational Monitoring
- 

## 3. Average Move Duration

### Purpose

Calculates average duration of cargo movement.

Average Move Duration =  
AVERAGE(  
fact\_cargo\_movements[move\_duration]  
)

Used In:

- Performance Analytics
  - Duration Monitoring
- 

## 4. Containers per Movement

### Purpose

Calculates average containers handled per movement.

Containers per Movement =  
DIVIDE(  
[Total Containers],  
[Total Movements]  
)

Used In:

- KPI Dashboard
-

## 5. Running Containers

### Purpose

Calculates cumulative container movement.

```
Running Containers =  
CALCULATE(  
[Total Containers],  
FILTER(  
ALL(dim_time),  
dim_time[date_id]  
<=  
MAX(dim_time[date_id])  
)  
)
```

Used In:

- Trend Analysis
  - Running Performance
- 

## 6. Containers YTD

### Purpose

Calculates Year-To-Date container total.

```
Containers YTD =  
TOTALYTD(  
[Total Containers],  
dim_time[date_id]  
)
```

Used In:

- Executive Dashboard
-

## 7. Previous Month Containers

### Purpose

Returns previous month's total containers.

Previous Month Containers =

```
CALCULATE(  
[Total Containers],  
DATEADD(  
dim_time[date_id],  
-1,  
MONTH  
)  
)
```

Used In:

- Growth Analysis
- 

## 8. Month-over-Month Growth %

### Purpose

Calculates monthly growth percentage.

MoM Growth % =

```
DIVIDE(  
[Total Containers]  
-  
[Previous Month Containers],  
[Previous Month Containers]  
)
```

Format:

Percentage

Used In:

- Performance Dashboard
- 

## 9. Terminal Rank

### Purpose

Ranks terminals based on container volume.

```
Terminal Rank =  
RANKX(  
ALL(  
dim_terminal[terminal_name]  
),  
[Total Containers],  
,  
DESC  
)
```

Used In:

- Terminal Ranking Table
- 

## 10. Maximum Move Duration

### Purpose

Returns maximum cargo duration.

```
Max Move Duration =  
MAX(  
fact_cargo_movements[move_duration]  
)
```

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# 11. Minimum Move Duration

## Purpose

Returns minimum cargo duration.

Min Move Duration =  
MIN(  
fact\_cargo\_movements[move\_duration]  
)

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# 12. Containers per Vessel

## Purpose

Calculates average containers per vessel.

Containers per Vessel =  
DIVIDE(  
[Total Containers],  
DISTINCTCOUNT(  
dim\_vessel[vessel\_name]  
)  
)

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## DAX Concepts Used

- SUM
- COUNT
- AVERAGE
- CALCULATE
- FILTER
- DIVIDE
- TOTALYTD
- DATEADD
- RANKX
- Time Intelligence

Developed using Microsoft Power BI Desktop